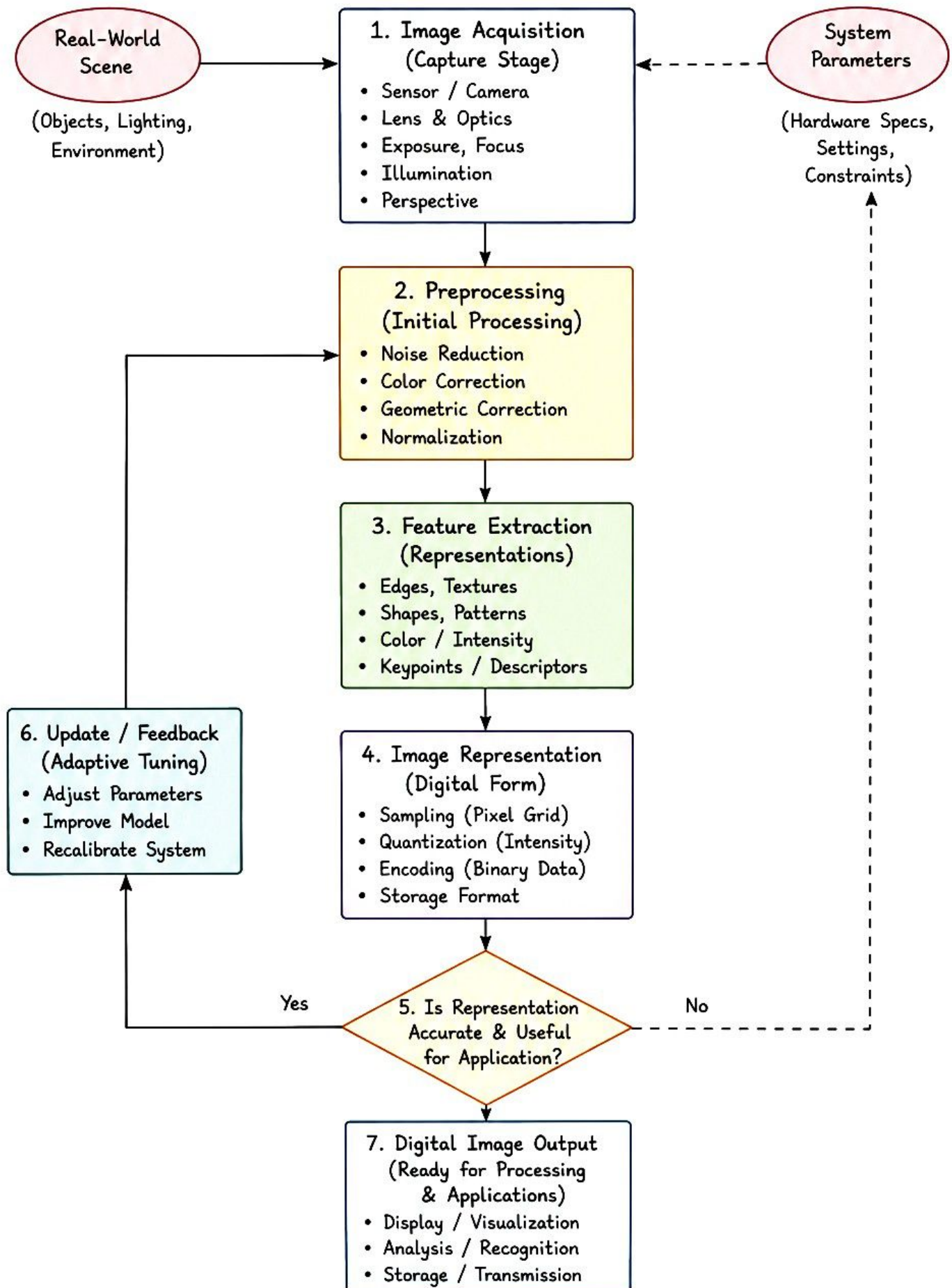


Q.No: 1

Pipeline from Real-World Scene Capture to Digital Image Representation



Legend:

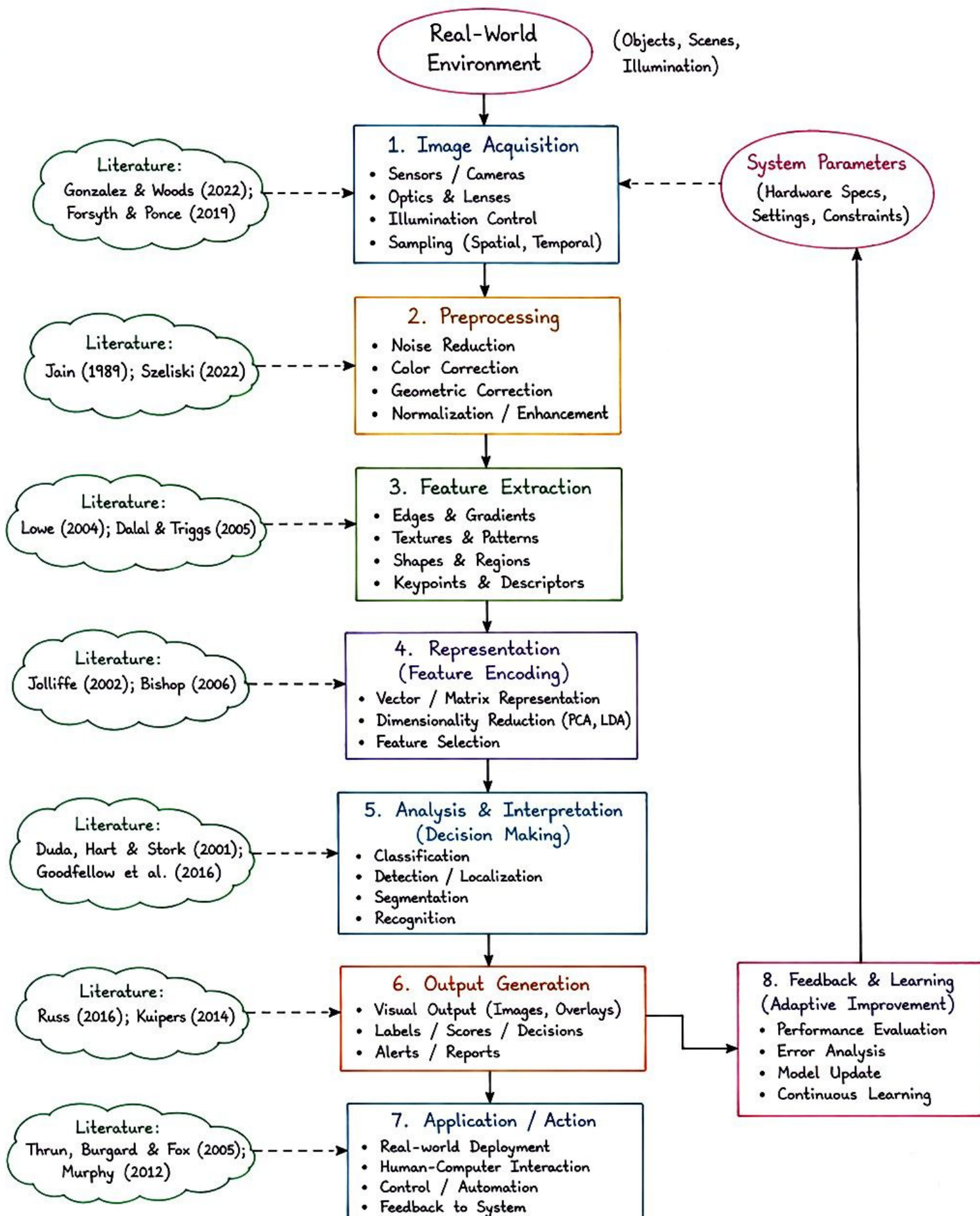
————> Main Flow

- - - -> Parameter Influence / Feedback

Description:

This architecture shows the complete pipeline from capturing a real-world scene to obtaining a digital image representation and using it for various applications, with feedback for continuous improvement.

Major Stages of Visual Data Handling in a Computer Vision System



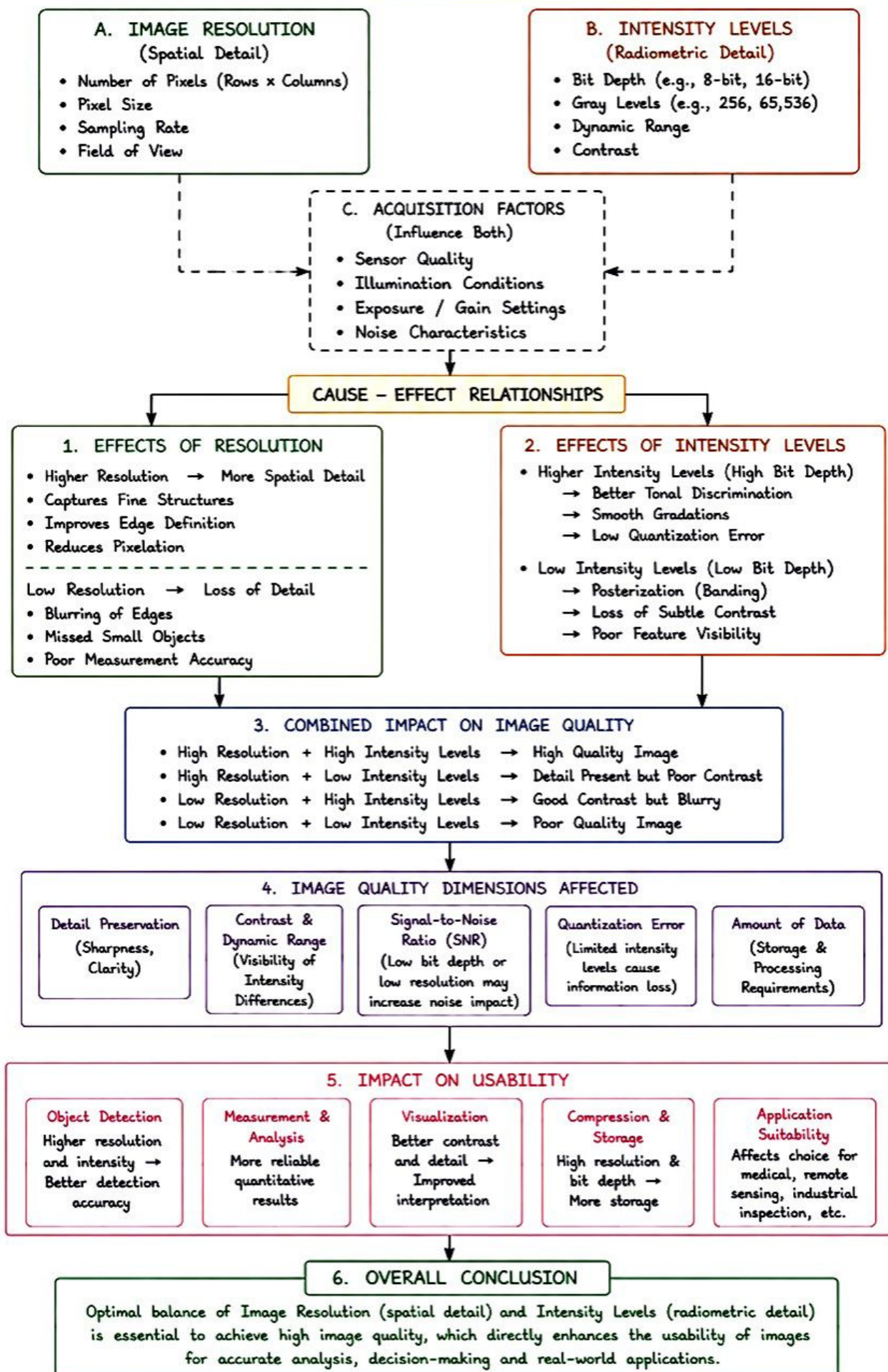
Legend:

- Main Flow
- - - - - Influence / Dependency
- ☁ Literature Connection

Description:

This map shows the hierarchical stages in a Computer Vision System from capturing visual data in the real world to producing meaningful outputs and improving the system through feedback.

Effect of Image Resolution and Intensity Levels on Image Quality and Usability

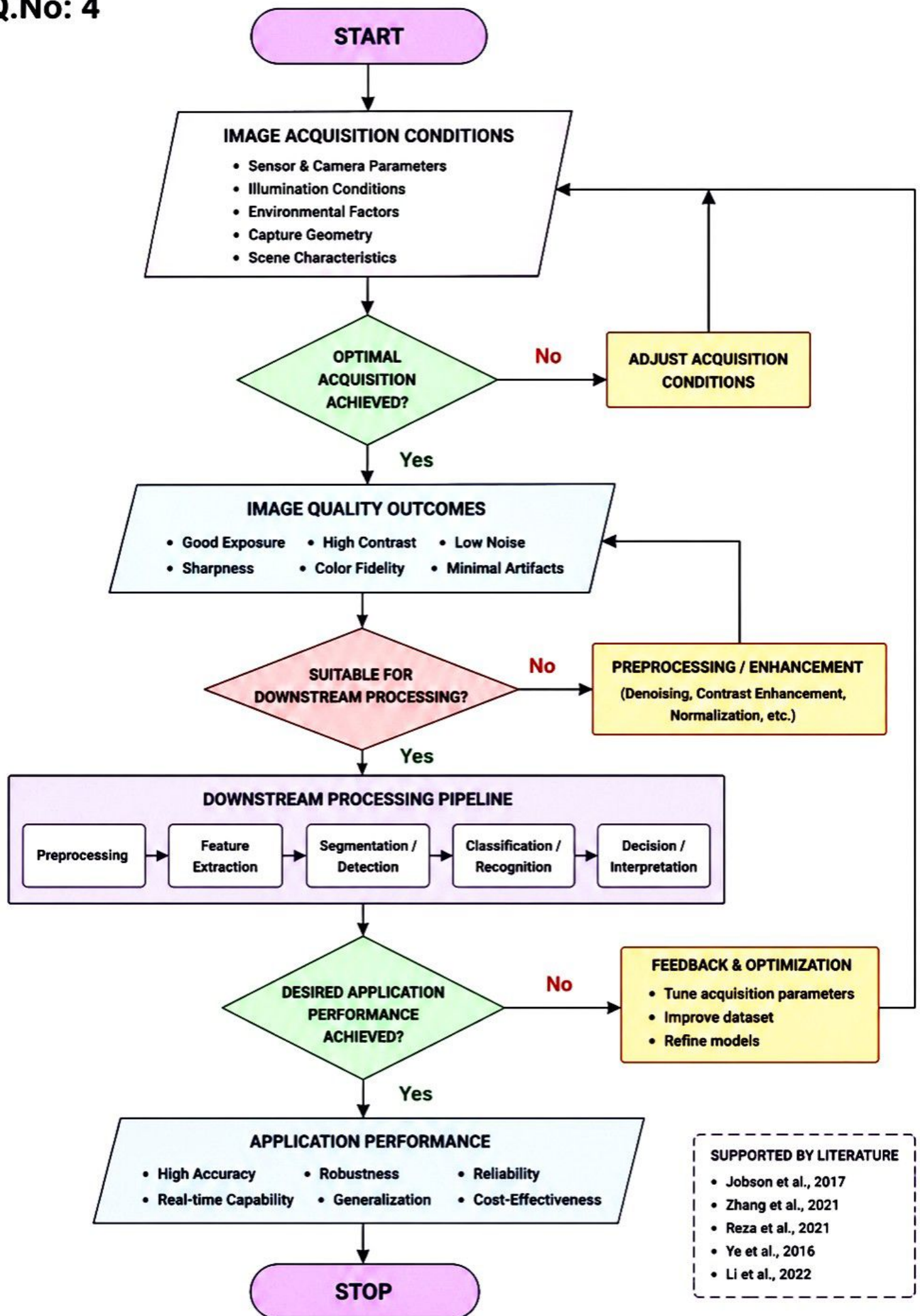


Literature Evidence

- Gonzalez, R. C., & Woods, R. E. (2018). Digital Image Processing (4th ed.). Pearson.
→ Higher resolution improves spatial detail; bit depth controls gray level resolution and contrast.
- Pratt, W. K. (2007). Digital Image Processing (4th ed.). Wiley-Interscience.
→ Quantization and sampling limitations degrade image fidelity and usability.

- Sonka, M., Hlavac, V., & Boyle, R. (2014). Image Processing, Analysis, and Machine Vision (4th ed.). Cengage.
→ Image quality depends on both spatial and radiometric resolution.
- Rangayyan, R. M. (2015). Biomedical Image Analysis (3rd ed.). CRC Press.
→ Appropriate resolution and intensity levels are critical for medical image diagnosis.

Q.No: 4



System Capabilities and Their Influence on Selection and Effectiveness of Vision-Based Applications

